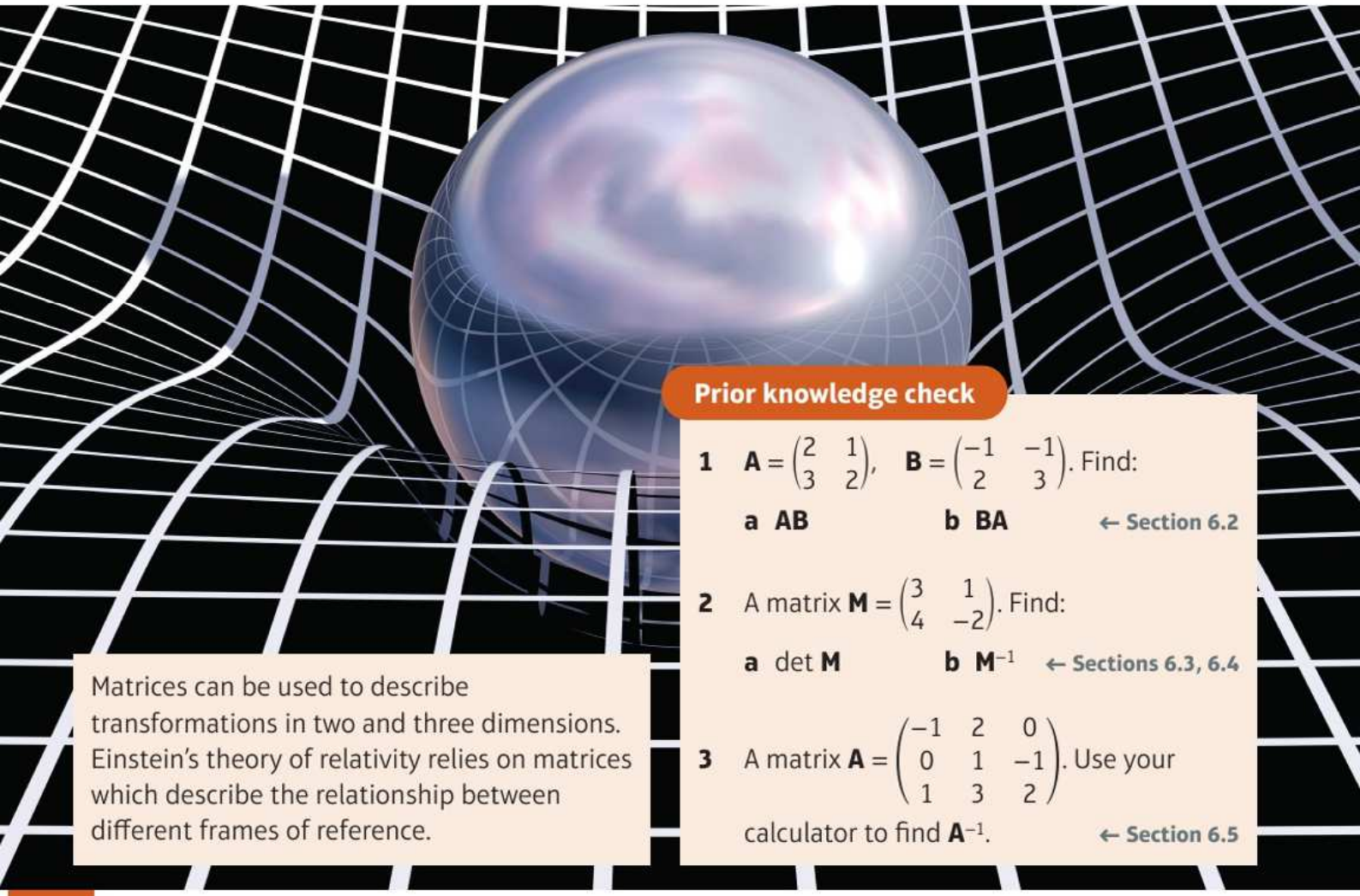
Linear transformations

7

Objectives

After completing this chapter you should be able to:

- Understand the properties of linear transformations and represent them using matrices → pages 127–130
- Perform reflections and rotations using matrices → pages 131–136
- Carry out enlargements and stretches using matrices → pages 136–140
- Find the coordinates of invariant points and the equations of invariant lines → pages 131–140
- Carry out successive transformations using matrix products → pages 140–144
- Understand linear transformations in three dimensions → pages 144–147
- Use inverse matrices to reverse linear transformations → pages 148–150



Matrices can be used to describe transformations in two and three dimensions. Einstein's theory of relativity relies on matrices which describe the relationship between different frames of reference.

Prior knowledge check

1 $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix}$, $\mathbf{B} = \begin{pmatrix} -1 & -1 \\ 2 & 3 \end{pmatrix}$. Find:

a **AB** **b** **BA** ← Section 6.2

2 A matrix $\mathbf{M} = \begin{pmatrix} 3 & 1 \\ 4 & -2 \end{pmatrix}$. Find:

a det **M** **b** $\mathbf{M}^{-1}$ ← Sections 6.3, 6.4

3 A matrix $\mathbf{A} = \begin{pmatrix} -1 & 2 & 0 \\ 0 & 1 & -1 \\ 1 & 3 & 2 \end{pmatrix}$. Use your calculator to find $\mathbf{A}^{-1}$. ← Section 6.5

7.1 Linear transformations in two dimensions

You can define a transformation in two dimensions by describing how a general point with position vector $\begin{pmatrix} x \\ y \end{pmatrix}$ is transformed. The new point is called the **image**.

Example 1

The three transformations S, T and U are defined as follows. Find the image of the point (2, 3) under each of these transformations.

$$S: \begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} x+4 \\ y-1 \end{pmatrix} \qquad T: \begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} 2x-y \\ x+y \end{pmatrix} \qquad U: \begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} 2y \\ -x^2 \end{pmatrix}$$

$S: \begin{pmatrix} 2 \\ 3 \end{pmatrix} \mapsto \begin{pmatrix} 2+4 \\ 3-1 \end{pmatrix} = \begin{pmatrix} 6 \\ 2 \end{pmatrix}$. The image is (6, 2)

Substitute $x = 2$ and $y = 3$ into the expressions for the image of the general point.

$T: \begin{pmatrix} 2 \\ 3 \end{pmatrix} \mapsto \begin{pmatrix} 2 \times 2 - 3 \\ 2+3 \end{pmatrix} = \begin{pmatrix} 1 \\ 5 \end{pmatrix}$. The image is (1, 5)

Substituting $x = 2$ and $y = 3$ into $2x - y$ gives 1 and into $x + y$ gives 5.

$U: \begin{pmatrix} 2 \\ 3 \end{pmatrix} \mapsto \begin{pmatrix} 2 \times 3 \\ -2^2 \end{pmatrix} = \begin{pmatrix} 6 \\ -4 \end{pmatrix}$. The image is (6, −4)

When $x = 2$, $-x^2 = -2^2 = -4$

A **linear transformation** has the special properties that the transformation only involves linear terms in x and y. In the example above, T is a linear transformation while S and U are not.

Watch out S represents a translation, but this is not a linear transformation since you can't write $x + 4$ in the form $ax + by$.

- **Linear transformations always map the origin onto itself.**
- **Any linear transformation can be represented by a matrix.**
- **The linear transformation** $T: \begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} ax+by \\ cx+dy \end{pmatrix}$ **can be represented by the matrix** $\mathbf{M} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ **since** $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} ax+by \\ cx+dy \end{pmatrix}$.

Note You can transform any point P by multiplying the transformation matrix by the position vector of P.

Example 2

Find matrices to represent these linear transformations.

a $T: \begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} 2y + x \\ 3x \end{pmatrix}$ **b** $V: \begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} -2y \\ 3x + y \end{pmatrix}$

a Transformation T is equivalent to

$T: \begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} 1x + 2y \\ 3x + 0y \end{pmatrix}$ — Write the transformation in the form $\begin{pmatrix} ax + by \\ cx + dy \end{pmatrix}$.

so the matrix is $\begin{pmatrix} 1 & 2 \\ 3 & 0 \end{pmatrix}$. — Use the coefficients of x and y to form the matrix.

b Transformation V is equivalent to

$V: \begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} 0x - 2y \\ 3x + y \end{pmatrix}$ — Write the transformation in the form $\begin{pmatrix} ax + by \\ cx + dy \end{pmatrix}$.

so the matrix is $\begin{pmatrix} 0 & -2 \\ 3 & 1 \end{pmatrix}$. — Use the coefficients of x and y to form the matrix.

Example 3

a The square S has coordinates (1, 1), (3, 1), (3, 3) and (1, 3). Find the vertices of the image of S under the transformation given by the matrix $\mathbf{M} = \begin{pmatrix} -1 & 2 \\ 2 & 1 \end{pmatrix}$.

b Sketch S and the image of S on a coordinate grid.

a $\begin{pmatrix} -1 & 2 \\ 2 & 1 \end{pmatrix}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$ — Write each point as a column vector and then use the usual rule for multiplying matrices.

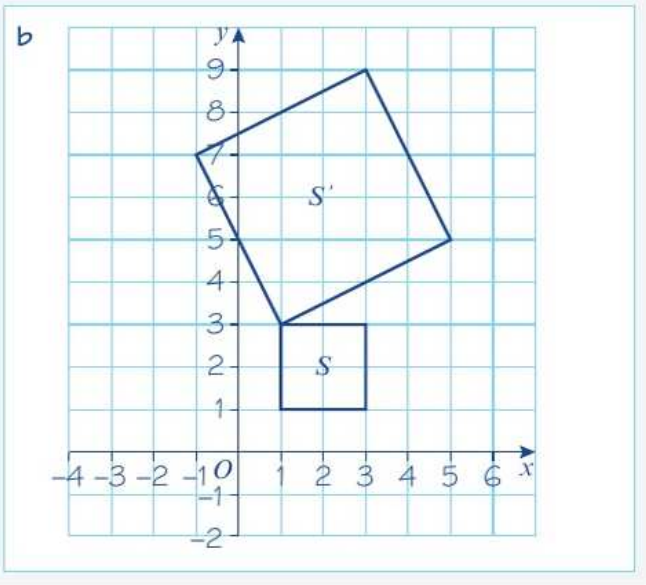
$\begin{pmatrix} -1 & 2 \\ 2 & 1 \end{pmatrix}\begin{pmatrix} 3 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 7 \end{pmatrix}$

$\begin{pmatrix} -1 & 2 \\ 2 & 1 \end{pmatrix}\begin{pmatrix} 3 \\ 3 \end{pmatrix} = \begin{pmatrix} 3 \\ 9 \end{pmatrix}$

$\begin{pmatrix} -1 & 2 \\ 2 & 1 \end{pmatrix}\begin{pmatrix} 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 5 \\ 5 \end{pmatrix}$

The vertices of the image of S, S', lie at (1, 3), (−1, 7), (3, 9) and (5, 5).

b



Notation S' is used to denote the image of S after a transformation.

Exercise 7A

1 Which of the following transformations are linear transformations?

a $P: \begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} 2x \\ y+1 \end{pmatrix}$ **b** $Q: \begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} x^2 \\ y \end{pmatrix}$ **c** $R: \begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} 2x+y \\ x+xy \end{pmatrix}$

d $S: \begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} y \\ -x \end{pmatrix}$ **e** $T: \begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} y+3 \\ x+3 \end{pmatrix}$ **f** $U: \begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} 2x \\ 3y-2x \end{pmatrix}$

2 Identify which of these are linear transformations and give their matrix representations. Give reasons to explain why the other transformations are not linear.

a $S: \begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} 2x-y \\ 3x \end{pmatrix}$ **b** $T: \begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} 2y+1 \\ x-1 \end{pmatrix}$ **c** $U: \begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} xy \\ 0 \end{pmatrix}$

d $V: \begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} 2y \\ -x \end{pmatrix}$ **e** $W: \begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} y \\ x \end{pmatrix}$

3 Identify which of these are linear transformations and give their matrix representations. Give reasons to explain why the other transformations are not linear.

a $S: \begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} x^2 \\ y^2 \end{pmatrix}$ **b** $T: \begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} -y \\ x \end{pmatrix}$ **c** $U: \begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} x-y \\ x-y \end{pmatrix}$

d $V: \begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ **e** $W: \begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} x \\ y \end{pmatrix}$

4 Find matrix representations for these linear transformations:

a $P: \begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} y+2x \\ -y \end{pmatrix}$ **b** $Q: \begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} -y \\ x+2y \end{pmatrix}$

5 The triangle T has vertices at $(-1, 1)$, $(2, 3)$ and $(5, 1)$.
Find the vertices of the image of T under the transformations represented by these matrices:

a $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$ **b** $\begin{pmatrix} 1 & 4 \\ 0 & -2 \end{pmatrix}$ **c** $\begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix}$

6 The square S has vertices at $(-1, 0)$, $(0, 1)$, $(1, 0)$ and $(0, -1)$.
Find the vertices of the image of S under the transformations represented by these matrices:

a $\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$ **b** $\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$ **c** $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

7 The rectangle R has vertices at $(2, 1)$, $(4, 1)$, $(4, 2)$ and $(2, 2)$.

a Find the vertices of the image of R under the transformation represented by the matrix $\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$.

b Sketch R and its image, R', on a coordinate grid.

c Describe fully the transformation that maps R onto R'.

8 A quadrilateral Q has coordinates $(1, 0)$, $(4, 2)$, $(3, 4)$ and $(0, 2)$.

a Find the vertices of the image of Q under the transformation represented by the matrix $\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$.

b Sketch Q and its image, Q', on a coordinate grid.

c Describe fully the transformation that maps Q onto Q'.

(P) **9** A square S has coordinates $(-1, 0)$, $(-3, 0)$, $(-3, 2)$ and $(-1, 2)$.

a Find the vertices of the image of S under the transformation represented by the matrix $\begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}$.

b Sketch S and the image of S on a coordinate grid.

c Describe fully the two transformations that map S onto S'.

10 A triangle T has vertices $(4, 1)$, $(4, 3)$ and $(1, 3)$.

a Find the vertices of the image of T under the transformation represented by the matrix $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$.

b Describe the effect of the transformation represented by the matrix $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$.

Challenge

A transformation T is defined as $T: \begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} 2x - 3y \\ x + y \end{pmatrix}$.

a Show that $T\begin{pmatrix} kx \\ ky \end{pmatrix} = kT\begin{pmatrix} x \\ y \end{pmatrix}$.

b Show that $T\left(\begin{pmatrix} x_1 \\ y_1 \end{pmatrix} + \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}\right) = T\begin{pmatrix} x_1 \\ y_1 \end{pmatrix} + T\begin{pmatrix} x_2 \\ y_2 \end{pmatrix}$.

Notation

$T\begin{pmatrix} x \\ y \end{pmatrix}$ is used to denote the image of the point $\begin{pmatrix} x \\ y \end{pmatrix}$ after the transformation T.

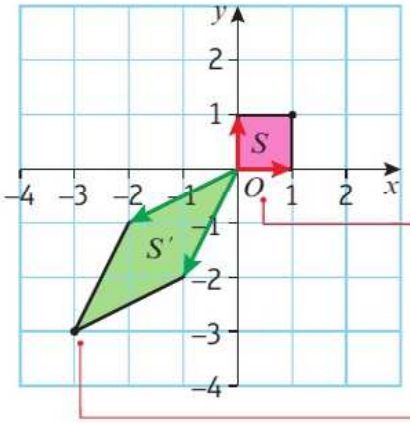
Note All linear transformations have these properties.

7.2 Reflections and rotations

Any linear transformation can be defined by the effect it has on unit vectors $\begin{pmatrix}1\\0\end{pmatrix}$ and $\begin{pmatrix}0\\1\end{pmatrix}$.

The transformation represented by the matrix $\mathbf{M} = \begin{pmatrix}a & b\\c & d\end{pmatrix}$ will map $\begin{pmatrix}1\\0\end{pmatrix}$ to $\begin{pmatrix}a\\c\end{pmatrix}$ and $\begin{pmatrix}0\\1\end{pmatrix}$ to $\begin{pmatrix}b\\d\end{pmatrix}$.

You can visualise this transformation by considering the unit square:



The linear transformation $\begin{pmatrix}-2 & -1\\-1 & -2\end{pmatrix}$ stretches and rotates the unit square S to produce the image S'.

The origin does not change under a linear transformation.

$T(\mathbf{a} + \mathbf{b}) = T\mathbf{a} + T\mathbf{b}$ so you can add the two image vectors to find the fourth vertex of the new shape. This means that the entire transformation can be defined by the images of the two unit vectors $\begin{pmatrix}1\\0\end{pmatrix}$ and $\begin{pmatrix}0\\1\end{pmatrix}$.

Points that do not move under the given transformation are called **invariant points**, and lines which do not move are called **invariant lines**.

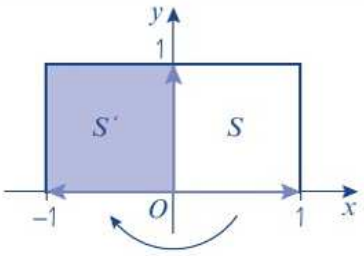
The only invariant point in the above transformation is the origin, which is always an invariant point of any linear transformation.

Example 4

The transformation U, represented by the 2×2 matrix $\mathbf{Q}$, is a reflection in the y-axis.

a Write down the matrix $\mathbf{Q}$.

b Write down the equation of the invariant line of this transformation.



Consider the unit square, and the effect that the transformation has on the unit vectors $\begin{pmatrix}1\\0\end{pmatrix}$ and $\begin{pmatrix}0\\1\end{pmatrix}$. This will completely define the transformation.

a $\begin{pmatrix}1\\0\end{pmatrix}$ maps to $\begin{pmatrix}-1\\0\end{pmatrix}$ and $\begin{pmatrix}0\\1\end{pmatrix}$ maps to $\begin{pmatrix}0\\1\end{pmatrix}$.

Hence the matrix $\mathbf{Q}$ is $\begin{pmatrix}-1 & 0\\0 & 1\end{pmatrix}$.

b The invariant line is the y-axis, so the equation of the invariant line is $x = 0$.

All of the points on the invariant line are invariant points.

- **A reflection in the y-axis is represented by the matrix $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$ and has the y-axis, which has equation $x = 0$, as an invariant line.**
- **A reflection in the x-axis is represented by the matrix $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ and has the x-axis, which has equation $y = 0$, as an invariant line.**

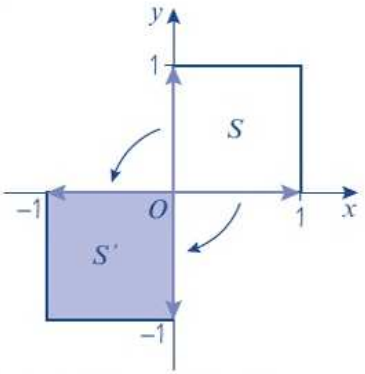
Example 5

$\mathbf{P} = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$

a Describe fully the single geometrical transformation U represented by the matrix $\mathbf{P}$.

b Given that U maps the point with coordinates (a, b) onto the point with coordinates $(3 + 2a, b + 1)$, find the values of a and b.

a $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ maps to $\begin{pmatrix} 0 \\ -1 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ maps to $\begin{pmatrix} -1 \\ 0 \end{pmatrix}$, hence the transformation U represented by matrix $\mathbf{P}$ is a reflection in the line $y = -x$.



You can visualise the transformation by sketching the effect it has on the unit square.

b $\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 3 + 2a \\ b + 1 \end{pmatrix}$

$\begin{pmatrix} -b \\ -a \end{pmatrix} = \begin{pmatrix} 3 + 2a \\ b + 1 \end{pmatrix}$

So $-b = 3 + 2a$ (1)

and $-a = b + 1$ (2)

Solving simultaneously gives $a = -2$ and $b = 1$.

Problem-solving

Write a matrix equation to show the transformation, then solve the resulting equations simultaneously to find the values of a and b.

You can solve two equations in two unknowns quickly using your calculator.

- **A reflection in the line $y = x$ is represented by the matrix $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and has invariant line with equation $y = x$.**
- **A reflection in the line $y = -x$ is represented by the matrix $\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$ and has invariant line with equation $y = -x$.**

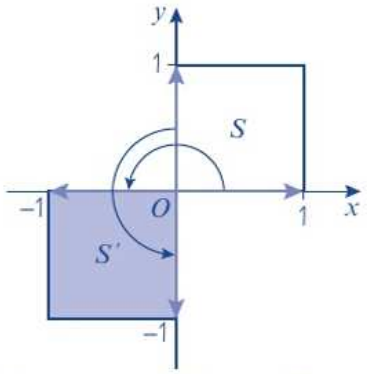
Example 6

The transformation U, represented by the 2×2 matrix $\mathbf{P}$, is a rotation of 180° about the point (0, 0).

a Write down the matrix $\mathbf{P}$.

b Write down the coordinates of the invariant point of this transformation.

a The given rotation is shown in the diagram:



$\begin{pmatrix}1\\0\end{pmatrix}$ maps to $\begin{pmatrix}-1\\0\end{pmatrix}$ and $\begin{pmatrix}0\\1\end{pmatrix}$ maps to $\begin{pmatrix}0\\-1\end{pmatrix}$.

Hence the matrix is $\begin{pmatrix}-1 & 0\\0 & -1\end{pmatrix}$.

b The only point that does not move under this transformation is the origin, so the coordinates of the invariant point are (0, 0).

If you need to find the matrix that represents a given transformation, it can help to draw a sketch transforming the unit square. Remember the transformation is defined by its effect on the unit vectors.

The origin is invariant under any linear transformation.

You need to be able to write down the matrix representing a rotation about any angle.

- **The matrix representing a rotation through angle θ anticlockwise about the origin is**

$$\begin{pmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{pmatrix}$$

The only invariant point is the origin (0, 0).

Note This general rotation matrix is given in the formulae booklet.

Example 7

$$\mathbf{M} = \begin{pmatrix}-\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2}\\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2}\end{pmatrix}$$

a Describe geometrically the rotation represented by $\mathbf{M}$.

b A square S has vertices at (1, 0), (2, 0), (2, 1) and (1, 1). Find the coordinates of the vertices of the image of S under the transformation described by $\mathbf{M}$.

a $\cos 135° = -\frac{\sqrt{2}}{2}$ and $\sin 135° = \frac{\sqrt{2}}{2}$ so **M** is a rotation, anticlockwise, through 135° about (0, 0).

b Apply the matrix to each vertex of S in turn:

$$\begin{pmatrix} -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{pmatrix}$$

$$\begin{pmatrix} -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{pmatrix}\begin{pmatrix} 2 \\ 0 \end{pmatrix} = \begin{pmatrix} -\sqrt{2} \\ \sqrt{2} \end{pmatrix}$$

$$\begin{pmatrix} -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{pmatrix}\begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} -\frac{3\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{pmatrix}$$

$$\begin{pmatrix} -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{pmatrix}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -\sqrt{2} \\ 0 \end{pmatrix}$$

The vertices of S' are $\left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$, $(-\sqrt{2}, \sqrt{2})$, $\left(-\frac{3\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$ and $(-\sqrt{2}, 0)$.

You are told that the **M** represents a rotation, so compare the matrix with $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$.

Online Explore rotations of the unit square using GeoGebra.



Work out the position vector of each vertex in the image of S.

Watch out Read the question carefully. Give your final answers as coordinates, not position vectors.

Exercise 7B

1 a Write down the matrix representing a reflection in the x-axis.
A triangle has vertices at $A = (1, 3)$, $B = (3, 3)$ and $C = (3, 2)$.

b Use matrices to show that the images of these vertices after a reflection in the x-axis are $A' = (1, -3)$, $B' = (3, -3)$ and $C' = (3, -2)$.

2 a Write down the matrix representing a reflection in the line $y = -x$.
A rectangle has vertices at $P = (1, 1)$, $Q = (1, 3)$, $R = (2, 3)$ and $S = (2, 1)$.

b Use matrices to show that the images of these vertices after a reflection in the line $y = -x$ are $P' = (-1, -1)$, $Q' = (-3, -1)$, $R' = (-3, -2)$ and $S' = (-1, -2)$.

3 Find the matrices that represent the following rotations.

a 90° anticlockwise about (0, 0)

b 270° anticlockwise about (0, 0)

c 45° anticlockwise about (0, 0)

d 210° anticlockwise about (0, 0)

e 135° clockwise about (0, 0)

Watch out The rotation matrix is for angles measured anticlockwise, so make sure you convert the clockwise angle to its equivalent anticlockwise angle.

4 A triangle has vertices at $A = (1, 1)$, $B = (4, 1)$ and $C = (4, 2)$. Find the exact coordinates of the vertices of the triangle after a rotation through:

a 90° anticlockwise about (0, 0)

b 150° anticlockwise about (0, 0)

5 A rectangle has vertices at $P = (2, 2)$, $Q = (2, 3)$, $R = (4, 3)$ and $S = (4, 2)$. Find the exact coordinates of the vertices of the rectangle after a rotation through:

a 270° anticlockwise about (0, 0)

b 135° clockwise about (0, 0)

E/P **6** $\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$

a Write down fully the transformations represented by the matrices **A** and **B**. **(4 marks)**

b The point (3, 2) is transformed by matrix **A**. Write down the coordinates of the image of this point. **(1 mark)**

c The point (a, b) is transformed onto the point $(a - 3b, 2a - 2b)$ by matrix **B**. Find the values of a and b. **(3 marks)**

E **7** $\mathbf{M} = \begin{pmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}$

a Write down fully the transformation represented by matrix **M**. **(2 marks)**

b The transformation represented by **M** maps the point (p, q) onto the point C with coordinates $(-\sqrt{2}, -2\sqrt{2})$. Find the values of p and q. **(4 marks)**

c Use your calculator to find $\mathbf{M}^3$. **(1 mark)**

d Point C is mapped onto the point D by the transformation represented by $\mathbf{M}^3$. Find the coordinates of point D and describe fully the transformation represented by $\mathbf{M}^3$. **(2 marks)**

E/P **8** **a** Describe fully the transformation represented by the matrix $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. **(2 marks)**

b Write down $\mathbf{A}^{50}$. **(1 mark)**

E/P **9** The matrix $\begin{pmatrix} a & b \\ c & -0.5 \end{pmatrix}$ represents an anticlockwise rotation about the origin, through an angle θ.

a Write down the value of a. **(1 mark)**

b Find two possible values of θ, and write down the matrix corresponding to each rotation. **(3 marks)**

E/P **10** **a** Write down the matrix representing a rotation through 270° clockwise about (0, 0). **(1 mark)**

b A point (a, b) transformed using this matrix is such that its image is the point $(a - 5b, 4b)$. Find the values of a and b. **(3 marks)**

Challenge

Prove that the general matrix representing a rotation through angle θ anticlockwise about the origin is $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$.

7.3 Enlargements and stretches

You can describe enlargements and stretches using linear transformations.

Example 8

$\mathbf{M} = \begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix}$

a Find the image T' of a triangle T with vertices (1, 1), (1, 2) and (2, 2) under the transformation represented by **M**.

b Sketch T and T' on the same set of coordinate axes.

c Describe geometrically the transformation represented by **M**.

a $\begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$

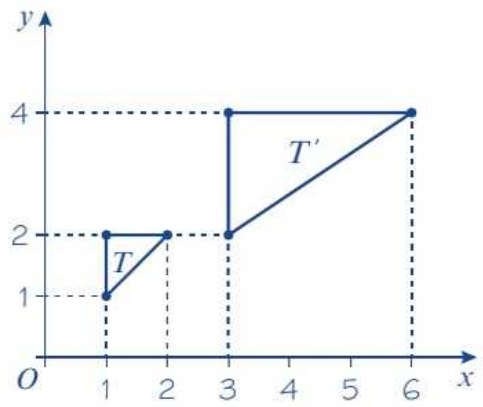
$\begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix}\begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$

$\begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix}\begin{pmatrix} 2 \\ 2 \end{pmatrix} = \begin{pmatrix} 6 \\ 4 \end{pmatrix}$

The coordinates of the image are (3, 2), (3, 4) and (6, 4).

Use matrix multiplication to find the image of each vertex under the transformation.

b



c The triangle has been stretched by a scale factor of 3 parallel to the x-axis and by a scale factor of 2 parallel to the y-axis.

Every x-coordinate has been multiplied by 3, and every y-coordinate has been doubled. Note that this is not the same as an enlargement, because the triangle has been stretched by different factors in the x- and y-directions.

Online Explore enlargements and stretches of triangle T using GeoGebra.



- **A transformation represented by the matrix $\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$ is a stretch of scale factor a parallel to the x-axis and a stretch of scale factor b parallel to the y-axis.**

  In the case where $a = b$, the transformation is an enlargement with scale factor a.

Note A stretch parallel to the x-axis only will have matrix $\begin{pmatrix} a & 0 \\ 0 & 1 \end{pmatrix}$.

A stretch parallel to the y-axis only will have matrix $\begin{pmatrix} 1 & 0 \\ 0 & b \end{pmatrix}$.

- **For a stretch parallel to the x-axis only, points on the y-axis are invariant. The line $x = 0$ is the invariant line.**

  For a stretch parallel to the y-axis only, points on the x-axis are invariant. The line $y = 0$ is the invariant line.

- **For stretches in both directions, including enlargements, there are no invariant lines and the only invariant point is the origin.**

Watch out Points on each axis map to **different** points on the same axis, so they are not invariant.

Reflections and rotations of 2D shapes both preserve the area of a shape. When a shape is stretched, its area can increase or decrease. You can use the determinant of the matrix representing this transformation to work out the scale factor for the change in area.

- **For a linear transformation represented by matrix M, det M represents the scale factor for the change in area. This is sometimes called the area scale factor.**

Watch out If the determinant of the matrix **M** is negative, the shape has been reflected.

Example 9

$\mathbf{M} = \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix}$

a Describe fully the transformation represented by **M**.

b A triangle T has vertices at (1, 0), (4, 0) and (4, 2). Find the area of the triangle.

c The triangle T is transformed using the matrix **M**. Use the determinant of **M** to find the area of the image of T.

a The matrix is of the form $\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$ so it is a stretch parallel to the x-axis, with scale factor 2 and a stretch parallel to the y-axis, with scale factor 4.

b Using $A = \frac{1}{2} \times \text{base} \times \text{height}$:

$A = \frac{1}{2} \times 3 \times 2 = 3$

c $\det \mathbf{M} = 8$

The area of the image of T is $3 \times 8 = 24$.

$\det \mathbf{M} = 2 \times 4 - 0 \times 0 = 8$ ← Section 6.3

The area increases by a factor of $\det \mathbf{M}$.

Exercise 7C

1 Write down the matrices representing the following linear transformations.

a A stretch with scale factor 4 parallel to the x-axis

b A stretch with scale factor 3 parallel to the y-axis

c An enlargement with scale factor 2

d A stretch with scale factor 5 parallel to the x-axis and a stretch scale factor $\frac{1}{2}$ parallel to the y-axis

2 For each of the transformations in question **1**, write down the area scale factor.

Hint In an enlargement, 'scale factor' refers to the linear scale factor of the enlargement so the area scale factor will be the square of this value.

3 The unit square is transformed using the matrix $\begin{pmatrix} 3 & 0 \\ 0 & 4 \end{pmatrix}$.

a Write down the coordinates of any invariant points.

b Work out the area of the resulting rectangle.

4 Write down the matrices representing the following transformations.

a A stretch with scale factor −2 parallel to the x-axis

b A stretch with scale factor −3 parallel to the x-axis and scale factor 4 parallel to the y-axis

c An enlargement with scale factor $-\frac{1}{2}$

E/P **5** $\mathbf{M} = \begin{pmatrix} 2 & 0 \\ 0 & -3 \end{pmatrix}$

a Describe fully the transformation represented by **M**. **(2 marks)**

A 2D shape with area k is transformed using the transformation represented by **M**.

b Given that the image of the shape has area 24, find the value of k. **(2 marks)**

6 A triangle with vertices at (1, 3), (5, 3) and (5, 2) is transformed using the matrix $\begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix}$.

a Find the coordinates of the vertices of the resulting image.

b Find the area of the new triangle.

7 A rectangle has vertices at (2, 0), (4, 0), (4, 5) and (2, 5). The rectangle is transformed by a stretch with scale factor 2 parallel to the x-axis and a stretch with scale factor −3 parallel to the y-axis.

a Find the coordinates of vertices of the resulting image.

b Find the area of the new rectangle.

E/P **8** $\mathbf{A} = \begin{pmatrix} 2\sqrt{5} & 0 \\ 0 & 2\sqrt{5} \end{pmatrix}$

a Describe fully the transformation represented by the matrix **A**. **(2 marks)**

b A triangle T has coordinates $(a, 1)$, (4, 1) and (4, 3). Given that T is transformed using matrix **A**, and the area of the resulting triangle is 60, find the value of a. **(3 marks)**

E/P **9** $\mathbf{M} = \begin{pmatrix} p & 1 \\ p & q \end{pmatrix}$ where p and q are constants and $q > 0$.

a Find $\mathbf{M}^2$ in terms of p and q. **(3 marks)**

Given that $\mathbf{M}^2$ represents an enlargement with centre (0, 0) and scale factor 6,

b find the values of p and q. **(3 marks)**

E/P **10** $\mathbf{A} = \begin{pmatrix} 5 & -1 \\ 2 & -2 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 2 & 1 \\ 2 & 5 \end{pmatrix}$

a Find the matrix **M** where **M** = **AB**. **(1 mark)**

b Describe fully the transformation represented by the matrix **M**. **(3 marks)**

c A triangle T has vertices at (2, 1), (6, 1) and (6, k). Given that T is transformed using matrix **M**, and the resulting triangle has area 320, find the value of k. **(4 marks)**

E **11** $\mathbf{M} = \begin{pmatrix} -1 & \sqrt{2} \\ -\sqrt{2} & -1 \end{pmatrix}$

A pentagon P of area 12 is transformed using matrix **M**. Find the area of the image of the pentagon P'.

(2 marks)

Hint You will learn how to describe transformations such as this in **Section 7.4**.

E/P **12** A triangle T has vertices at the points $A = (k, 1)$, $B = (4, 1)$ and $C = (4, k)$ where k is an integer constant.

Triangle T is transformed by the matrix $\begin{pmatrix} 4 & -1 \\ k & 2 \end{pmatrix}$.

Given that triangle T has a right angle at B, and the area of the image triangle T' is 10, find the value of k. **(5 marks)**

E **13** A triangle T has vertices at (0, 0), (7, 7) and (3, −2).

a Write down the matrix, **M**, which represents a rotation through 45° anticlockwise about (0, 0). **(1 mark)**

b Find the exact coordinates of the image of T when T is transformed using **M**. **(3 marks)**

c Show that $\det \mathbf{M} = 1$. **(2 marks)**

d Hence find the area of the original triangle T. **(1 mark)**

Challenge

A transformation U is represented by the matrix $\mathbf{P} = \begin{pmatrix} 7 & 0 \\ 0 & 0 \end{pmatrix}$.

a Find $\det \mathbf{P}$.

b Show that any point in the xy-plane is mapped onto the x-axis by U.

Problem-solving

A non-zero singular 2 × 2 matrix maps any point in the plane onto a straight line through the origin.

7.4 Successive transformations

You can use matrix products to represent combinations of transformations.

- **The matrix PQ represents the transformation Q, with matrix Q followed by the transformation P, with matrix P.**

Example 10

a Find the 2 × 2 matrix **T** that represents a rotation through 90° anticlockwise about the origin followed by a reflection in the line $y = x$.

b Describe the single transformation represented by **T**.

a Rotation 90° anticlockwise about the origin: $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$

Reflection in the line $y = x$: $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

$\mathbf{T} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

b **T** represents reflection in the line $y = 0$.

Write down the two matrices for the single transformations.

Find the matrix product to determine the single matrix that has the same effect as these combined transformations. Make sure you apply the matrix product in the right order.

Check where **T** maps the position vectors $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$.

Example 11

$$\mathbf{M} = \begin{pmatrix} -2\sqrt{2} & -2\sqrt{2} \\ 2\sqrt{2} & -2\sqrt{2} \end{pmatrix}$$

The matrix **M** represents an enlargement with scale factor k followed by a rotation through angle θ anticlockwise about the origin.

a Find the value of k.

b Find the value of θ.

a $\det \mathbf{M} = 16$

Area scale factor = 16

$k = \sqrt{16} = 4$

b $\begin{pmatrix} -2\sqrt{2} & -2\sqrt{2} \\ 2\sqrt{2} & -2\sqrt{2} \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}\begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix}$

$= \begin{pmatrix} 4\cos\theta & -4\sin\theta \\ 4\sin\theta & 4\cos\theta \end{pmatrix}$

$4\cos\theta = -2\sqrt{2}$

$\cos\theta = \frac{-2\sqrt{2}}{4} = -\frac{\sqrt{2}}{2}$

$\theta = 135°$ or $\theta = 225°$

Check using the lower-left element:

$\sin\theta = \frac{\sqrt{2}}{2}$ so $\theta = 135°$

Use your calculator to find det **M**.

k is the linear scale factor. The rotation does not affect area, so $k = \sqrt{\det \mathbf{M}}$.

If the enlargement matrix is **Q** and the rotation matrix is **P** then **M** = **PQ**.

Use one element to find possible values of θ.

Watch out You will need to use one of the sin elements to check which angle is correct. $\sin\theta$ is positive so choose the angle in the second quadrant.

Exercise 7D

1 $\mathbf{A} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$, $\mathbf{B} = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$, $\mathbf{C} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$

Find these matrix products and describe the single transformation represented by each product.

a **AB** **b** **BA** **c** **AC** **d** $\mathbf{A}^2$ **e** $\mathbf{C}^2$

2 A = rotation of 90° anticlockwise about (0, 0) B = rotation of 180° about (0, 0)

C = reflection in the x-axis D = reflection in the y-axis

a Find matrix representations of each of the four transformations A, B, C and D.

b Use matrix products to identify the single geometric transformation represented by each of these combinations.

i Reflection in the x-axis followed by a rotation of 180° about (0, 0)

ii Rotation of 180° about (0, 0) followed by a reflection in the x-axis

iii Reflection in the y-axis followed by reflection in the x-axis

iv Reflection in the y-axis followed by rotation of 90° anticlockwise about (0, 0)

v Rotation of 180° about (0, 0) followed by a second rotation of 180° about (0, 0)

vi Reflection in the x-axis followed by rotation of 90° anticlockwise about (0, 0) followed by a reflection in the y-axis

vii Reflection in the y-axis followed by rotation of 180° about (0, 0) followed by a reflection in the x-axis

3 $\mathbf{R} = \begin{pmatrix} 3 & 0 \\ 0 & -2 \end{pmatrix}$, $\mathbf{S} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ and $\mathbf{T} = \begin{pmatrix} 5 & 0 \\ 0 & 5 \end{pmatrix}$

Find these matrix products and, where possible, use your knowledge of the standard forms of transformation matrices to find the single transformation represented by the products:

a **RS** **b** **RT** **c** **TS** **d** **TR** **e** **ST** **f** **RST**

4 A is a stretch with scale factor 2 parallel to the x-axis, and scale factor 3 parallel to the y-axis.

B is an enlargement with scale factor −2.

C is an enlargement with scale factor 4.

a Write down the matrices representing each of the transformations A, B and C.

b Find the single 2 × 2 matrix representing each of the following combined transformations:

i B followed by A **ii** C followed by A

iii B followed by C **iv** C followed by C

v C followed by B followed by A

Hint If C is represented by matrix **M**, then C followed by C will be represented by $\mathbf{M}^2$.

(P) **5** Use matrices to show that a refection in the y-axis followed by a reflection in the line $y = -x$ is equivalent to a rotation of 90° anticlockwise about (0, 0).

(E/P) **6** A student makes the following claim:

> If T is a reflection in the x-axis and U is a rotation 90° anticlockwise about the origin then T followed by U is the same as U followed by T.

Show, using matrix multiplication, that the student is incorrect. **(4 marks)**

(E/P) **7** $\mathbf{P} = \begin{pmatrix} -4 & 0 \\ 0 & 2 \end{pmatrix}$ and $\mathbf{Q} = \begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix}$, where k is a constant.

a Find the matrix product **PQ**. **(2 marks)**

b Describe, in terms of k, the single transformation represented by **PQ**. **(2 marks)**

c Show that for any value of k, **PQ** = **QP**. **(2 marks)**

(E/P) **8** $\mathbf{A} = \begin{pmatrix} 3 & 0 \\ 0 & 4 \end{pmatrix}$

a Find the matrix $\mathbf{A}^2$. **(1 mark)**

b Describe fully the transformation represented by $\mathbf{A}^2$. **(2 marks)**

$\mathbf{B} = \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$ where a and b are constants.

c Find the general matrix $\mathbf{B}^2$ and state, in terms of a and b, the transformation represented by this matrix. **(3 marks)**

E/P **9** The matrix **R** is given by $\begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$

a Find $\mathbf{R}^2$. **(1 mark)**

b Describe the geometric transformation represented by $\mathbf{R}^2$. **(2 marks)**

c Hence describe the geometric transformation represented by **R**. **(1 marks)**

d Write down $\mathbf{R}^8$. **(1 mark)**

E/P **10** $\mathbf{M} = \begin{pmatrix} -\frac{3}{\sqrt{2}} & \frac{3}{\sqrt{2}} \\ -\frac{3}{\sqrt{2}} & -\frac{3}{\sqrt{2}} \end{pmatrix}$

The matrix **M** represents an enlargement with scale factor k ($k < 0$) followed by a rotation of angle θ anticlockwise about the origin.

a Find the value of k. **(2 marks)**

b Find the value of θ. **(3 marks)**

E/P **11** $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 5 & 0 \\ 0 & 5 \end{pmatrix}$

A triangle T is transformed using matrix **B**. The image is then transformed using matrix **A**. Given that the area of the image, T' is 75, find the area of T. **(3 marks)**

E/P **12** The transformation T is a rotation through 225° anticlockwise about the origin.

a Write down the matrix representing this transformation. **(1 mark)**

The transformation U is a reflection in the line $y = x$.

b Write down the matrix representing this transformation. **(1 mark)**

c Find the matrix representing the combined transformation of U followed by T. **(2 marks)**

E/P **13** $\mathbf{A} = \begin{pmatrix} k & \sqrt{3} \\ \sqrt{3} & -k \end{pmatrix}$ where k is a constant.

a Find, in terms of k, the matrix $\mathbf{A}^2$. **(2 marks)**

b Describe fully the transformation represented by $\mathbf{A}^2$. **(2 marks)**

E/P **14** $\mathbf{P} = \begin{pmatrix} a & b \\ b & -a \end{pmatrix}$ where a and b are constants.

Show that the general matrix $\mathbf{P}^2$ represents an enlargement, and write down, in terms of a and b, the scale factor of the enlargement. **(3 marks)**

Challenge

a Given that $\mathbf{P} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$, show algebraically that $\mathbf{P}^2 = \begin{pmatrix} \cos 2\theta & -\sin 2\theta \\ \sin 2\theta & \cos 2\theta \end{pmatrix}$.

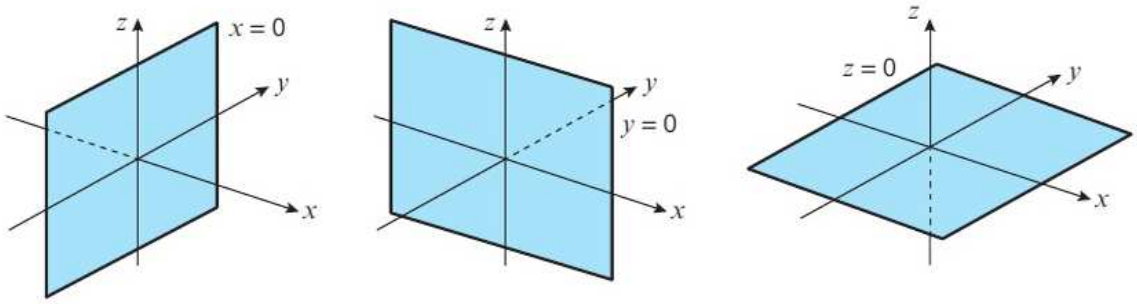
b Interpret this result geometrically.

7.5 Linear transformations in three dimensions

Any linear transformation in three dimensions can be defined by the effect it has on the unit vectors $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$. The transformation represented by the matrix $\mathbf{M} = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$ will map $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ to $\begin{pmatrix} a \\ d \\ g \end{pmatrix}$, $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ to $\begin{pmatrix} b \\ e \\ h \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ to $\begin{pmatrix} c \\ f \\ i \end{pmatrix}$.

You need to be able to carry out transformations in three dimensions that are reflections in the planes $x = 0$, $y = 0$ or $z = 0$.

Links In three dimensions the coordinate axes are labelled x, y and z. ← **Pure Year 2, Chapter 12**



Example 12

A transformation U, in three dimensions, represents a reflection in the plane $z = 0$.

a Write down the 3 × 3 matrix that represents this transformation.

b Find the image of the point $(-1, 2, 3)$ under this transformation.

a The point with position vector $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ lies in the xy-plane so stays where it is.

The point with position vector $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ lies in the xy-plane so stays where it is.

The plane $z = 0$ contains the x- and y-axes. Hence points in the xy-plane are invariant.

The point with position vector $\begin{pmatrix}0\\0\\1\end{pmatrix}$ reflects in the xy-plane to the point with position vector $\begin{pmatrix}0\\0\\-1\end{pmatrix}$.

Hence the matrix representing the transformation is $\begin{pmatrix}1&0&0\\0&1&0\\0&0&-1\end{pmatrix}$.

b $\begin{pmatrix}1&0&0\\0&1&0\\0&0&-1\end{pmatrix}\begin{pmatrix}-1\\2\\3\end{pmatrix}=\begin{pmatrix}-1\\2\\-3\end{pmatrix}$

The coordinates of the image are $(-1, 2, -3)$.

The x- and y-coordinates stay the same, and the sign of the z-coordinate changes.

- **A reflection in the plane $x = 0$ is represented by the matrix $\begin{pmatrix}-1&0&0\\0&1&0\\0&0&1\end{pmatrix}$.**
- **A reflection in the plane $y = 0$ is represented by the matrix $\begin{pmatrix}1&0&0\\0&-1&0\\0&0&1\end{pmatrix}$.**
- **A reflection in the plane $z = 0$ is represented by the matrix $\begin{pmatrix}1&0&0\\0&1&0\\0&0&-1\end{pmatrix}$.**

You also need to be able to carry out rotations about one of the coordinate axes in three dimensions.

- **A rotation, angle θ, about the x-axis is represented by the matrix $\begin{pmatrix}1&0&0\\0&\cos\theta&-\sin\theta\\0&\sin\theta&\cos\theta\end{pmatrix}$.**
- **A rotation, angle θ, about the y-axis is represented by the matrix $\begin{pmatrix}\cos\theta&0&\sin\theta\\0&1&0\\-\sin\theta&0&\cos\theta\end{pmatrix}$.**
- **A rotation, angle θ, about the z-axis is represented by the matrix $\begin{pmatrix}\cos\theta&-\sin\theta&0\\\sin\theta&\cos\theta&0\\0&0&1\end{pmatrix}$.**

Online Explore rotations about the coordinate axes using GeoGebra.



Note In all cases, θ is the angle measured anticlockwise when facing in the positive direction.

Example 13

$$\mathbf{M} = \begin{pmatrix} \frac{\sqrt{3}}{2} & 0 & \frac{1}{2} \\ 0 & 1 & 0 \\ -\frac{1}{2} & 0 & \frac{\sqrt{3}}{2} \end{pmatrix}$$

a Describe the transformation represented by **M**.

b Find the image of the point with coordinates (−1, −2, 1) under the transformation represented by **M**.

a Rotation about the y-axis.

State the axis of rotation. You need to be familiar with the general forms of matrices for rotations about each of the coordinate axes.

$\cos\theta = \frac{\sqrt{3}}{2}$ so $\theta = 30°$ or $330°$

Use the top-left element to determine two possible angles.

$\sin\theta = \frac{1}{2}$ so $\theta = 30°$

$\sin\theta$ is positive so choose the angle in the first quadrant.

The transformation is a rotation anticlockwise about the y-axis through an angle of 30°.

b
$$\begin{pmatrix} \frac{\sqrt{3}}{2} & 0 & \frac{1}{2} \\ 0 & 1 & 0 \\ -\frac{1}{2} & 0 & \frac{\sqrt{3}}{2} \end{pmatrix} \begin{pmatrix} -1 \\ -2 \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{1-\sqrt{3}}{2} \\ -2 \\ \frac{1+\sqrt{3}}{2} \end{pmatrix}$$

Use your calculator to find this matrix product.

The coordinates of the image are

$$\left(\frac{1-\sqrt{3}}{2}, -2, \frac{1+\sqrt{3}}{2}\right)$$

Exercise 7E

1 Write down the matrices representing the following transformations.

a Reflection in the plane $x = 0$

b Reflection in the plane $y = 0$

c Rotation of 180° about the y-axis

d Rotation of 90° anticlockwise about the z-axis

e Rotation of 270° anticlockwise about the y-axis

f Rotation of 300° anticlockwise about the x-axis

2 Describe the transformations represented by the following matrices.

a $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$ **b** $\begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{pmatrix}$ **c** $\begin{pmatrix} -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & 0 \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix}$

E/P **3** $\mathbf{M} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}$

a Determine the single transformation represented by the matrix **M**. **(3 marks)**

b The point $A = (3, -1, 4)$ is transformed using this matrix. Find the coordinates of the image of A. **(1 mark)**

c The point $B = (a, -a, 2a - 1)$ is transformed to the point with coordinates $(a, a - 5, -a)$ using matrix **M**. Find the value of a. **(3 marks)**

E/P **4** **P** is the matrix representing a rotation of 120° anticlockwise about the z-axis.

a Write down the matrix **P**. **(1 mark)**

b A point $Q = (3, -1, 0)$ is transformed using the matrix **P**. Find the coordinates of the image of Q. **(1 mark)**

c A point $R = (k, 0, k)$ is transformed using matrix **P**. Find, in terms of k, the exact coordinates of the image of R. **(3 marks)**

E **5** **A** is the matrix representing a reflection in the plane $x = 0$ and **B** is the matrix representing a reflection in plane $y = 0$.

a Write down the matrices **A** and **B**. **(2 marks)**

b The point $P(a, b, c)$ is transformed using matrix **A**. Find the coordinates of P' in terms of a, b and c. **(2 marks)**

c P' is transformed using matrix **B**. Find the coordinates of the image of P' in terms of a, b and c. **(2 marks)**

E **6** $\mathbf{M} = \begin{pmatrix} -\frac{\sqrt{3}}{2} & 0 & -\frac{1}{2} \\ 0 & 1 & 0 \\ \frac{1}{2} & 0 & -\frac{\sqrt{3}}{2} \end{pmatrix}$

a Find the transformation represented by matrix **M**. **(3 marks)**

b The point with coordinates $(k, -k, 0)$ is transformed using matrix **M**. Find, in terms of k, the exact coordinates of the image of this point. **(3 marks)**

P **7** **a** Write down the matrix representing a rotation of 315° anticlockwise about the y-axis.

A tetrahedron T has vertices at $(1, 0, -1)$, $(1, 1, -1)$, $(3, 2, 3)$ and $(0, 0, 0)$.

b Find the images of the vertices of the tetrahedron under the transformation described in part **a**.

c Hence find the volume of T.

Hint The formula for the volume of a tetrahedron is $\frac{1}{3} \times$ base area $\times$ height.

Challenge

a Find the 3 × 3 matrix representing a reflection in the plane $x = 0$ followed by a reflection in the plane $y = 0$.

b Find the 3 × 3 matrix representing a rotation of 45° anticlockwise about the x-axis followed by a reflection in the plane $x = 0$.

7.6 The inverse of a linear transformation

Since $\mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$, you can use inverse matrices to reverse the effect of a linear transformation.

- **The transformation described by the matrix $\mathbf{A}^{-1}$ has the effect of reversing the transformation described by the matrix A.**

Links Reflections are **self-inverse**. ← Section 6.4, 6.5

Example 14

The matrix $\mathbf{A} = \begin{pmatrix} 2 & 4 \\ -2 & -5 \end{pmatrix}$ represents a transformation T.

Given that T maps point P with coordinates (x, y) onto the point P' with coordinates (6, 10),

a find the coordinates of P.

The matrix **B** represents a transformation U. Given that the transformation T followed by the transformation U is equivalent to a reflection in the line $y = x$,

b find **B**.

a $\mathbf{A}^{-1} = \begin{pmatrix} 2.5 & 2 \\ -1 & -1 \end{pmatrix}$

$\begin{pmatrix} 2 & 4 \\ -2 & -5 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 6 \\ 10 \end{pmatrix}$

$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2.5 & 2 \\ -1 & -1 \end{pmatrix}\begin{pmatrix} 6 \\ 10 \end{pmatrix}$

$= \begin{pmatrix} 35 \\ -16 \end{pmatrix}$

P has coordinates (35, −16).

This represents the inverse transformation to T. You can find it quickly using your calculator.

$\mathbf{A}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 6 \\ 10 \end{pmatrix}$

Left multiply both sides by $\mathbf{A}^{-1}$:

$\mathbf{A}^{-1}\mathbf{A}\begin{pmatrix} x \\ y \end{pmatrix} = \mathbf{A}^{-1}\begin{pmatrix} 6 \\ 10 \end{pmatrix}$ so $\begin{pmatrix} x \\ y \end{pmatrix} = \mathbf{A}^{-1}\begin{pmatrix} 6 \\ 10 \end{pmatrix}$

b $\mathbf{BA} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

$\mathbf{B} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 2.5 & 2 \\ -1 & -1 \end{pmatrix} = \begin{pmatrix} -1 & -1 \\ 2.5 & 2 \end{pmatrix}$

The matrix representing T followed by U is **BA**. This is equal to $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, which is the matrix for a reflection in the line $y = x$.

Right multiply both sides by $\mathbf{A}^{-1}$. Remember that the order is important.

$(\mathbf{BA})\mathbf{A}^{-1} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\mathbf{A}^{-1}$, so $\mathbf{B} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\mathbf{A}^{-1}$

Exercise 7F

1 The matrix $\mathbf{R} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$

a Give a geometrical interpretation of the transformation represented by **R**.

b Find $\mathbf{R}^{-1}$.

c Give a geometrical interpretation of the transformation represented by $\mathbf{R}^{-1}$.

2 a The matrix $\mathbf{S} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$

i Give a geometrical interpretation of the transformation represented by **S**.

ii Show that $\mathbf{S}^2 = \mathbf{I}$.

iii Give a geometrical interpretation of the transformation represented by $\mathbf{S}^{-1}$.

b The matrix $\mathbf{T} = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$

i Give a geometrical interpretation of the transformation represented by **T**.

ii Show that $\mathbf{T}^2 = \mathbf{I}$.

iii Give a geometrical interpretation of the transformation represented by $\mathbf{T}^{-1}$.

c Calculate det **S** and det **T** and comment on their values in the light of the transformations they represent.

3 The matrix **A** represents a reflection in the line $y = x$ and the matrix **B** represents an anticlockwise rotation of 270° about (0, 0).

a Find the matrix $\mathbf{C} = \mathbf{BA}$ and interpret it geometrically.

b Find $\mathbf{C}^{-1}$ and give a geometrical interpretation of the transformation represented by $\mathbf{C}^{-1}$.

c Find the matrix $\mathbf{D} = \mathbf{AB}$ and interpret it geometrically.

d Find $\mathbf{D}^{-1}$ and give a geometrical interpretation of the transformation represented by $\mathbf{D}^{-1}$.

E **4** The matrix $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ -3 & -2 \end{pmatrix}$ represents a transformation T.

Given that T maps point P with coordinates (x, y) onto the point P' with coordinates (5, 8),

a find the coordinates of P. **(2 marks)**

The matrix **B** represents a transformation U. Given that the transformation T followed by the transformation U is equivalent to a reflection in the line $y = -x$,

b find **B**. **(2 marks)**

E **5** $\mathbf{E} = \begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix}$

a State the transformation represented by matrix **E**. **(1 mark)**

b Use your calculator to find $\mathbf{E}^{-1}$. **(1 mark)**

A triangle T transformed using matrix **E** is such that the coordinates of the image are (4, 6), (9, 7) and (3, 1).

c Using your answer to part **b**, find the coordinates of the vertices of T. **(2 marks)**

E/P **6** $\mathbf{M} = \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$ where a and b are non-zero constants.

a Find, in terms of a and b, the matrix $\mathbf{M}^{-1}$. **(2 marks)**

b The point $D = (p, q)$ maps to the point $(-6, 8)$ under the transformation represented by **M**. Find, in terms of a and b, the coordinates of D. **(3 marks)**

E **7** $\mathbf{R} = \begin{pmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}$

a Describe fully the transformation represented by **R**. **(2 marks)**

b The point (p, q) is mapped onto $\left(\frac{1-2\sqrt{3}}{2}, \frac{2+\sqrt{3}}{2}\right)$ under the transformation represented by **R**. Find the values of p and q. **(3 marks)**

E/P **8** $\mathbf{A} = \begin{pmatrix} 2 & 4 \\ 1 & 3 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$

Triangle T is transformed to triangle T' using the matrix **AB**.
Find the matrix **P** such that triangle T' is mapped to T. **(3 marks)**

E/P **9** $\mathbf{A} = \begin{pmatrix} 6 & -2 \\ -4 & 1 \end{pmatrix}$

The transformation represented by **A** maps the point P onto the point Q.
Given that Q has coordinates (a, b), find the coordinates of P in terms of a and b. **(4 marks)**

E **10** $\mathbf{M} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & -2 \\ 1 & 3 & -1 \end{pmatrix}$

The point (a, b, c) maps to the point $(3, 2, -1)$ under **M**. Find $\mathbf{M}^{-1}$ and hence find the exact values of a, b and c. **(4 marks)**

E/P **11** The 3×3 matrix **P** represents a reflection in the plane $x = 0$. The matrix **Q** represents an anticlockwise rotation through 90° about the z-axis.

a Find the matrix $\mathbf{M} = \mathbf{PQ}$ and interpret it geometrically. **(3 marks)**

b Use your calculator to find $\mathbf{M}^{-1}$ and give a geometrical interpretation. **(2 marks)**

c Find the matrix $\mathbf{N} = \mathbf{QP}$ and interpret it geometrically. **(3 marks)**

d Show that $\mathbf{N}^{-1} = \mathbf{P}^{-1}\mathbf{Q}^{-1}$ and interpret geometrically the matrix $\mathbf{N}^{-1}$. **(3 marks)**

E/P **12** The matrix $\mathbf{A} = \begin{pmatrix} 1 & 3 & -1 \\ -1 & 1 & 2 \\ 2 & 0 & -1 \end{pmatrix}$ represents a transformation T, in three dimensions.

The matrix **B** represents a transformation U. Given that transformation T followed by transformation U represents a reflection in the xy-plane, find **B**. **(4 marks)**

Mixed exercise 7

E/P **1** The matrix **Y** represents an anticlockwise rotation of 90° about (0, 0).

a Find **Y**. **(1 mark)**

The matrices **A** and **B** are such that **AB** = **Y**. Given that $\mathbf{B} = \begin{pmatrix} 3 & 2 \\ 2 & 1 \end{pmatrix}$,

b find **A**. **(3 marks)**

c Describe the transformation represented by the matrix **ABABABAB** and write down its 2 × 2 matrix. **(2 marks)**

E **2** The matrix **R** represents a reflection in the x-axis and the matrix **E** represents an enlargement with scale factor 2 and centre (0, 0).

a Find the matrix **C** = **ER** and give a geometrical interpretation of the transformation **C** represents. **(4 marks)**

b Find $\mathbf{C}^{-1}$ and give a geometrical interpretation of the transformation represented by $\mathbf{C}^{-1}$. **(2 marks)**

E/P **3** $\mathbf{P} = \begin{pmatrix} 0 & k \\ k & 0 \end{pmatrix}$

Given that **P** represents a transformation T, followed by a reflection in the line $y = x$,

a find, in terms of k, a matrix representing T. **(4 marks)**

b Given that the point (−3, −2) maps to point (9, 6) under T, find the value of k. **(2 marks)**

E/P **4** The matrix $\mathbf{M} = \begin{pmatrix} 2\sqrt{3} & -2 \\ 2 & 2\sqrt{3} \end{pmatrix}$ represents a rotation followed by an enlargement.

a Find the scale factor of the enlargement. **(2 marks)**

b Find the angle of rotation. **(3 marks)**

A point P is mapped onto a point P' under **M**. Given that the coordinates of P' are (a, b),

c find, in terms of a and b, the coordinates of P. **(4 marks)**

E **5** $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$

a Describe fully the transformations represented by the matrices **A** and **B**. **(4 marks)**

b The point (p, q) is transformed by the matrix product **AB**. Give the coordinates of the image of this point in terms of p and q. **(2 marks)**

E/P **6** $\mathbf{M} = \begin{pmatrix} -4 & 3 \\ 1 & -2 \end{pmatrix}$

A triangle T has vertices at $(k, 2)$, (6, 2) and (6, 7). Given that T is transformed using matrix **M**, and the area of the resulting triangle is 110, find the two possible values of k. **(3 marks)**

E **7** $\mathbf{A} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 4 & 0 \\ 0 & 3 \end{pmatrix}$

a Find the matrix **P** = **AB**. **(1 mark)**

A triangle T is transformed using matrix **P**.

b Given that the area of T' is 60, find the area of T. **(2 marks)**

E/P **8** $\mathbf{A} = \begin{pmatrix} -\frac{\sqrt{3}}{2} & -\frac{1}{2} & 0 \\ \frac{1}{2} & -\frac{\sqrt{3}}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix}$

a Find the transformation represented by matrix **A**. **(3 marks)**

b The point with coordinates $(a, b, -a)$ is transformed using matrix **M**. Find, in terms of a and b, the exact coordinates of the image of this point. **(3 marks)**

E/P **9** $\mathbf{P} = \begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix}$ where a is a non-zero constant.

a Find, in terms of a, the matrix $\mathbf{P}^{-1}$. **(2 marks)**

b The point A maps to the point (4, 7) under the transformation represented by **M**. Find, in terms of a, the coordinates of A. **(3 marks)**

E/P **10** The matrix $\mathbf{P} = \begin{pmatrix} -1 & 2 \\ -5 & 8 \end{pmatrix}$ represents a transformation U.

A triangle T is transformed by transformation U followed by an anticlockwise rotation through 90° about the origin. The resulting image is labelled T'.

Find a matrix **M** representing a linear transformation that maps T' back onto T. **(3 marks)**

E **11** $\mathbf{A} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 4 & -1 \\ 3 & -2 \end{pmatrix}$

The transformation represented by **B** followed by the transformation represented by **A** is equivalent to the transformation represented by matrix **P**.

a Find **P**. **(1 mark)**

Triangle T is transformed to the triangle T' by the transformation represented by **P**.

Given that the area of the triangle T' is 35,

b find the area of triangle T. **(3 marks)**

Triangle T' is transformed to the original triangle T by the transformation represented by matrix **Q**.

c Find **Q**. **(2 marks)**

E **12** $\mathbf{M} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ 0 & -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{pmatrix}$

The point (a, b, c) maps to the point (0, 1, 1) under **M**. Find $\mathbf{M}^{-1}$ and hence find the exact values of a, b and c. **(4 marks)**

E/P **13** The matrix $\mathbf{A} = \begin{pmatrix} 1 & -1 & 0 \\ 2 & 0 & 4 \\ 3 & -2 & -1 \end{pmatrix}$ represents a transformation T.

Given that T maps point P with coordinates (x, y, z) onto the point P' with coordinates $(3, 7, 4)$,

a find the coordinates of P. **(3 marks)**

The matrix **B** represents a transformation U. Given that the transformation T followed by the transformation U is equivalent to a rotation through 90° about the x-axis,

b find **B**. **(3 marks)**

Challenge

1 Find the 3 × 3 matrix representing the single transformation that is equivalent to a reflection in the plane $y = 0$, followed by a rotation of 90° about the x-axis, followed by a reflection in the plane $z = 0$.

2 a Show that the transformation described by $\begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix}$ maps any point in the plane onto the line $y = x$.

b Find the matrix representing the linear transformation that maps any point in the plane onto the straight line $y = mx$.

c Explain why, in general, the transformation that maps any point in the plane onto the straight line $ax + by = c$ is **not** linear.

Summary of key points

1 • Linear transformations always map the origin onto itself.
• Any linear transformation can be represented by a matrix.

2 The linear transformation $T: \begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} ax + by \\ cx + dy \end{pmatrix}$ can be represented by the matrix $\mathbf{M} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ since $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} ax + by \\ cx + dy \end{pmatrix}$.

3 A reflection in the y-axis is represented by the matrix $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$ and has the y-axis, which has equation $x = 0$, as an invariant line.

4 A reflection in the x-axis is represented by the matrix $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ and has the x-axis, which has equation $y = 0$, as an invariant line.

5 A reflection in the line $y = x$ is represented by the matrix $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and has invariant line with equation $y = x$.

6 A reflection in the line $y = -x$ is represented by the matrix $\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$ and has invariant line with equation $y = -x$.

7 The matrix representing a rotation through angle θ anticlockwise about the origin is $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$.
The only invariant point is the origin (0, 0).

8 A transformation represented by the matrix $\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$ is a stretch of scale factor a parallel to the x-axis and a stretch of scale factor b parallel to the y-axis.
In the case where $a = b$, the transformation is an enlargement with scale factor a.

9 For a stretch parallel to the x-axis only, points on the y-axis are invariant. The line $x = 0$ is the invariant line.

10 For a stretch parallel to the y-axis only, points on the x-axis are invariant. The line $y = 0$ is the invariant line.

11 For stretches in both directions, including enlargements, there are no invariant lines and the only invariant point is the origin.

12 For a linear transformation represented by matrix **M**, det **M** represents the scale factor for the change in area. This is sometimes called the **area scale factor**.

13 The matrix **PQ** represents the transformation Q, with matrix **Q**, followed by the transformation P, with matrix **P**.

14 A reflection in the plane $x = 0$ is represented by the matrix $\begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$.

15 A reflection in the plane $y = 0$ is represented by the matrix $\begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$.

16 A reflection in the plane $z = 0$ is represented by the matrix $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$.

17 A rotation, angle θ, anticlockwise about the x-axis is represented by the matrix

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{pmatrix}$$

18 A rotation, angle θ, anticlockwise about the y-axis is represented by the matrix

$$\begin{pmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{pmatrix}$$

19 A rotation, angle θ, anticlockwise about the z-axis is represented by the matrix

$$\begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

20 The transformation described by the matrix $\mathbf{A}^{-1}$ has the effect of reversing the transformation described by the matrix **A**.